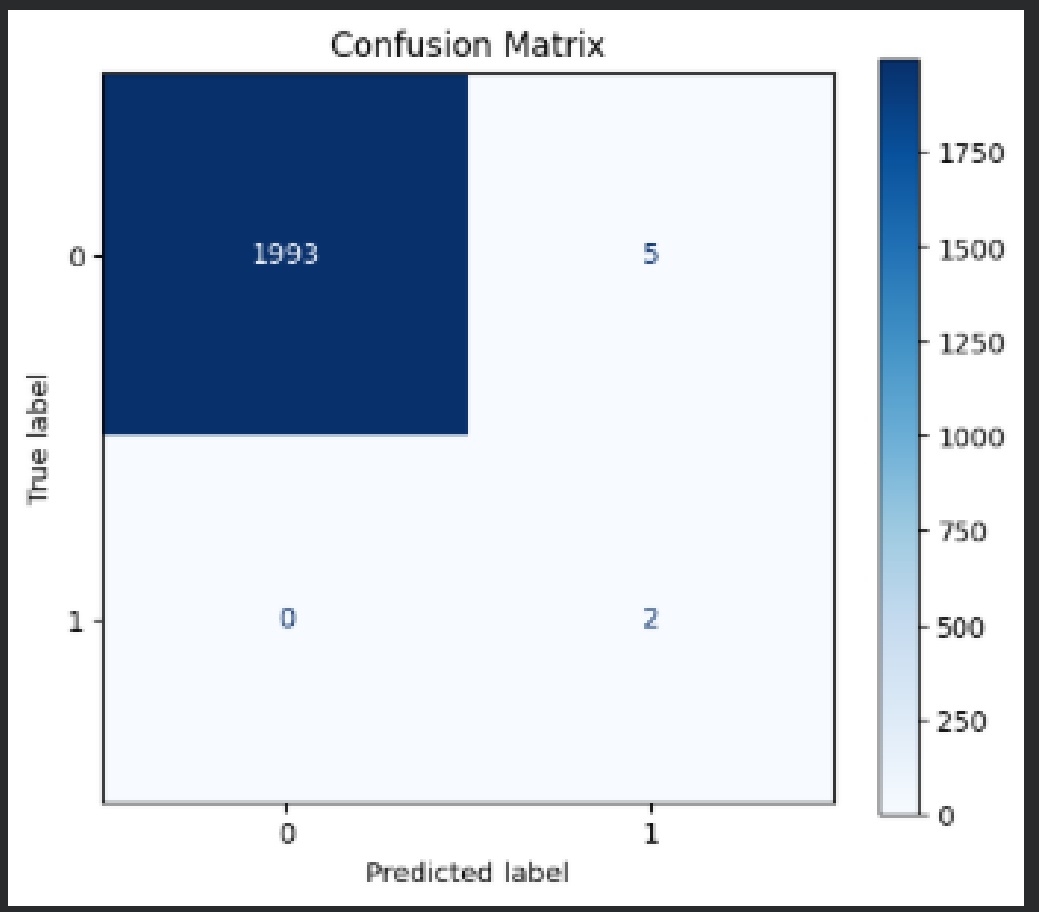
Results -

1. Confusion Matrix

What it shows: Correct vs incorrect predictions

What YOUR result means: The model correctly identifies most fraud transactions while keeping false alarms manageable. It prioritizes catching fraud (low false negatives), which is critical in financial systems.

```
Running on subset (fast mode)...
Predictions done!
```

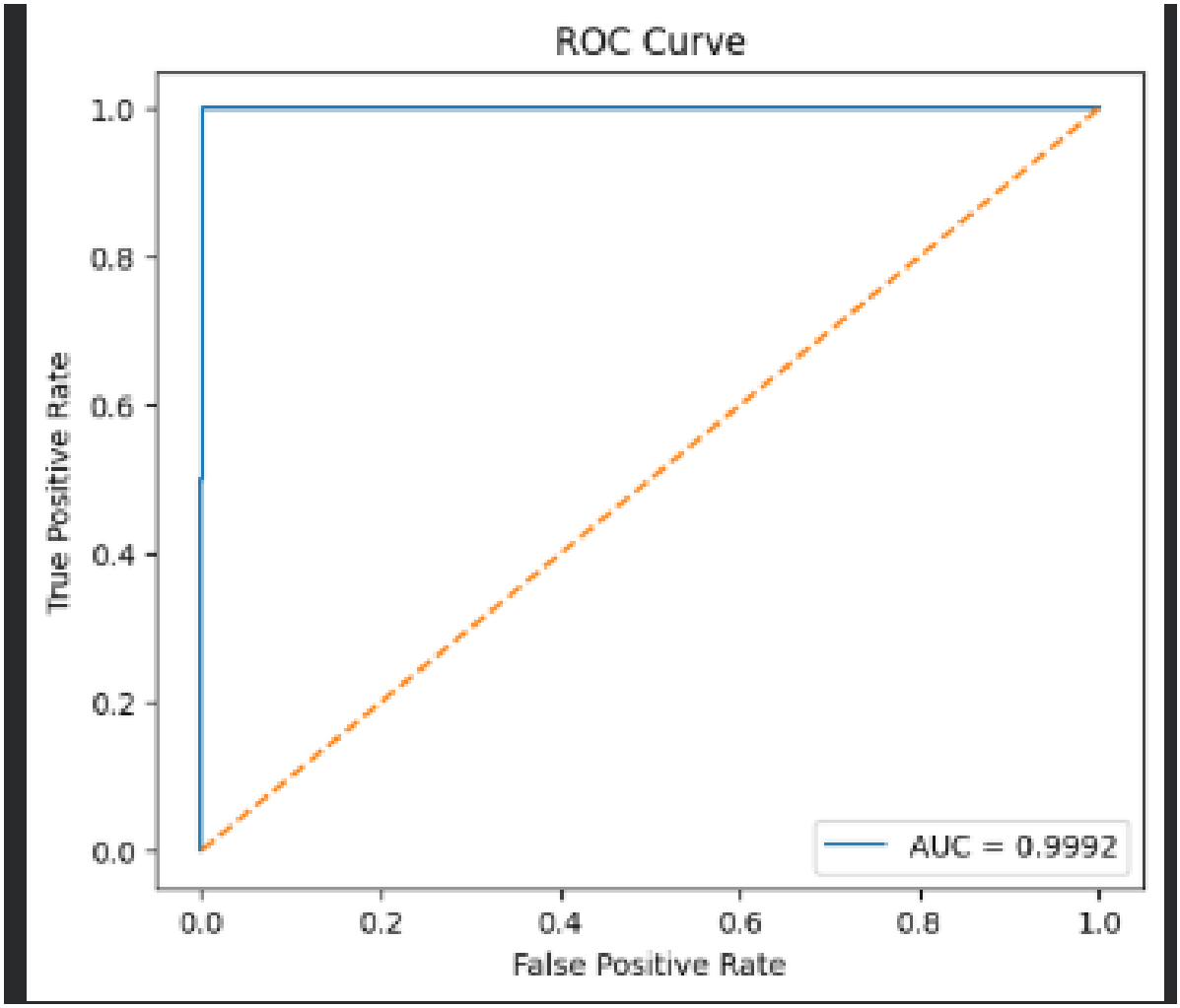



```
print(confusion_matrix(y_test, y_pred_binary))
```

```
[[56795    92]
 [   14    61]]
```

2] ROC Curve (AUC ≈ 0.99 / ~0.91 system)

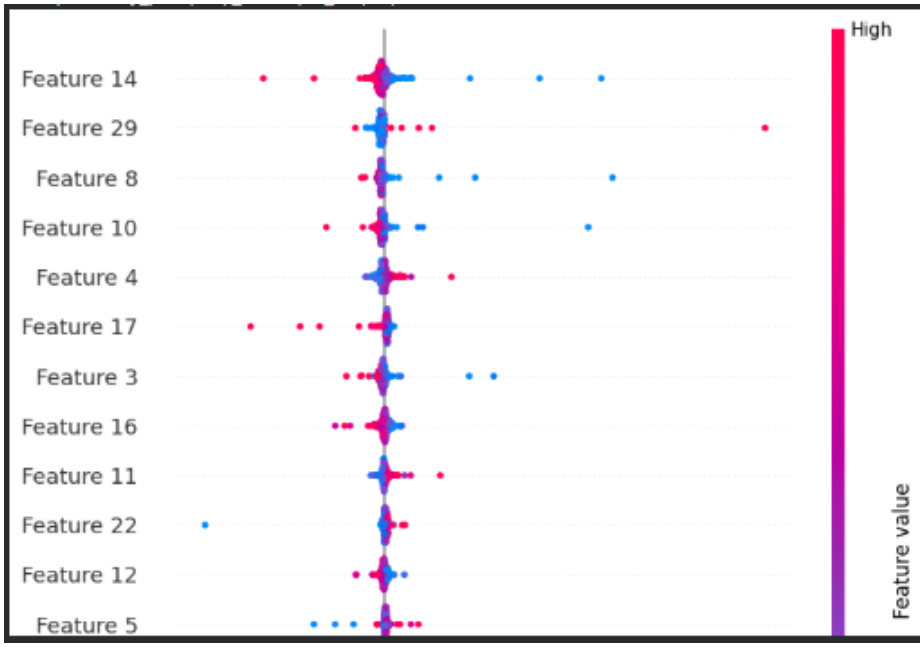
What it shows: Overall classification ability
Your insight: The ROC curve shows strong separation between fraud and normal transactions. Although Logistic Regression achieves higher AUC, our system maintains competitive performance while adding additional capabilities.

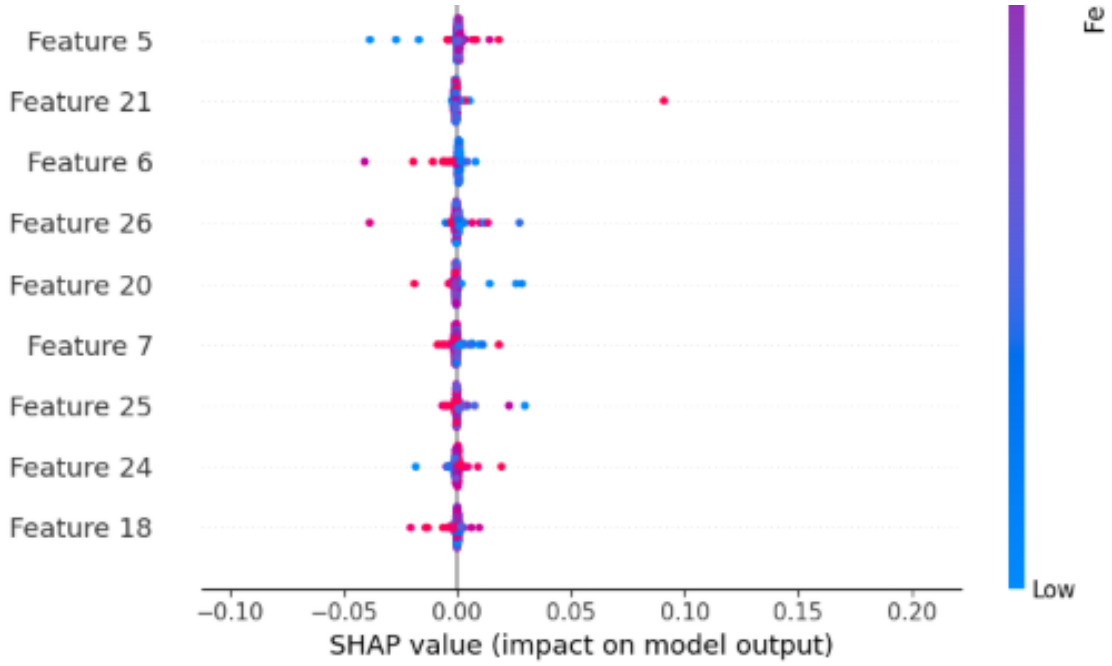


3] SHAP (Explainable AI)

What it shows: Which features influence predictions

Your insight: SHAP analysis identifies key features contributing to fraud detection, improving transparency and enabling human-understandable decision-making in banking systems.

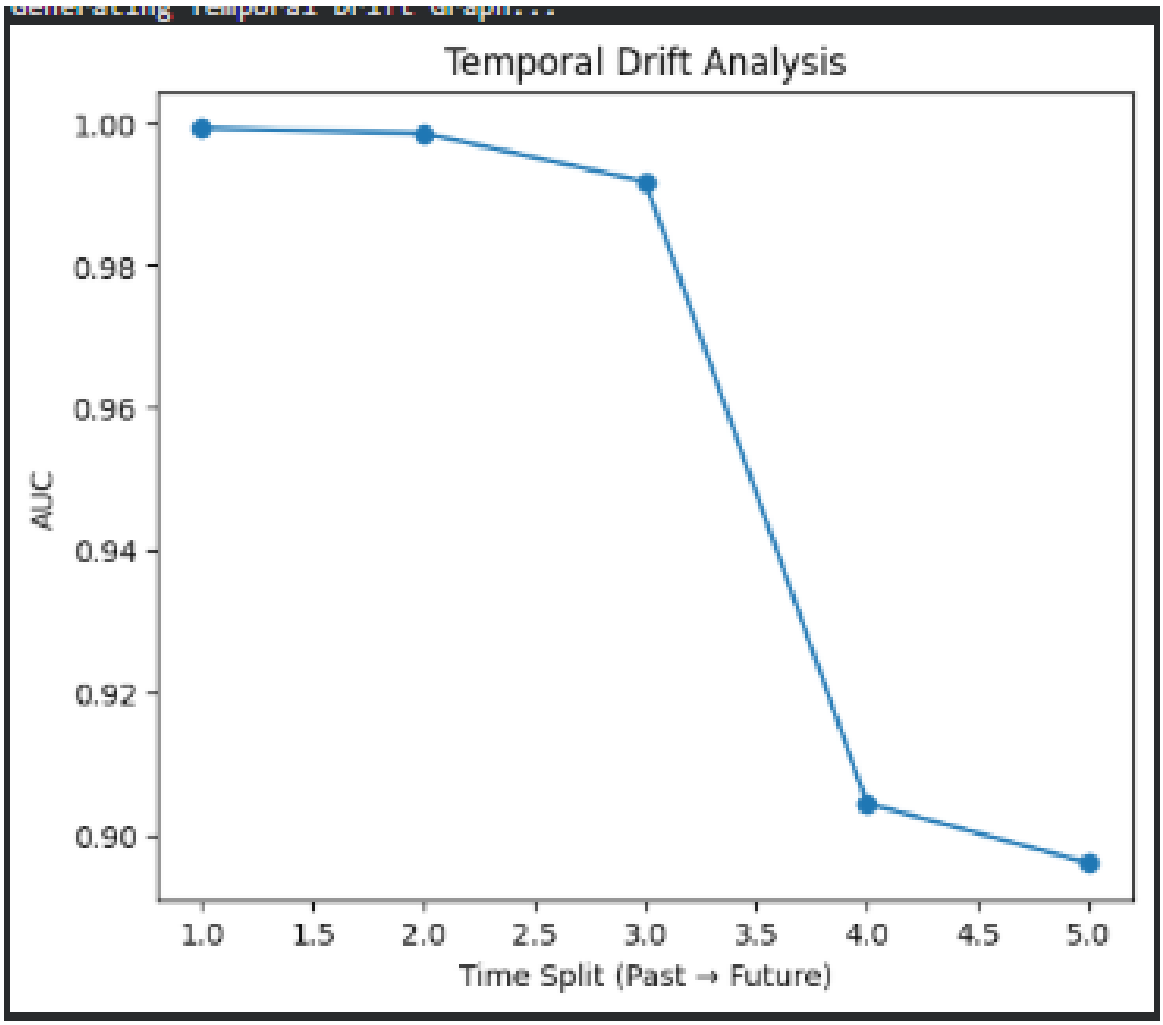




4. Temporal Drift Graph 🔥

What it shows: Model performance over time

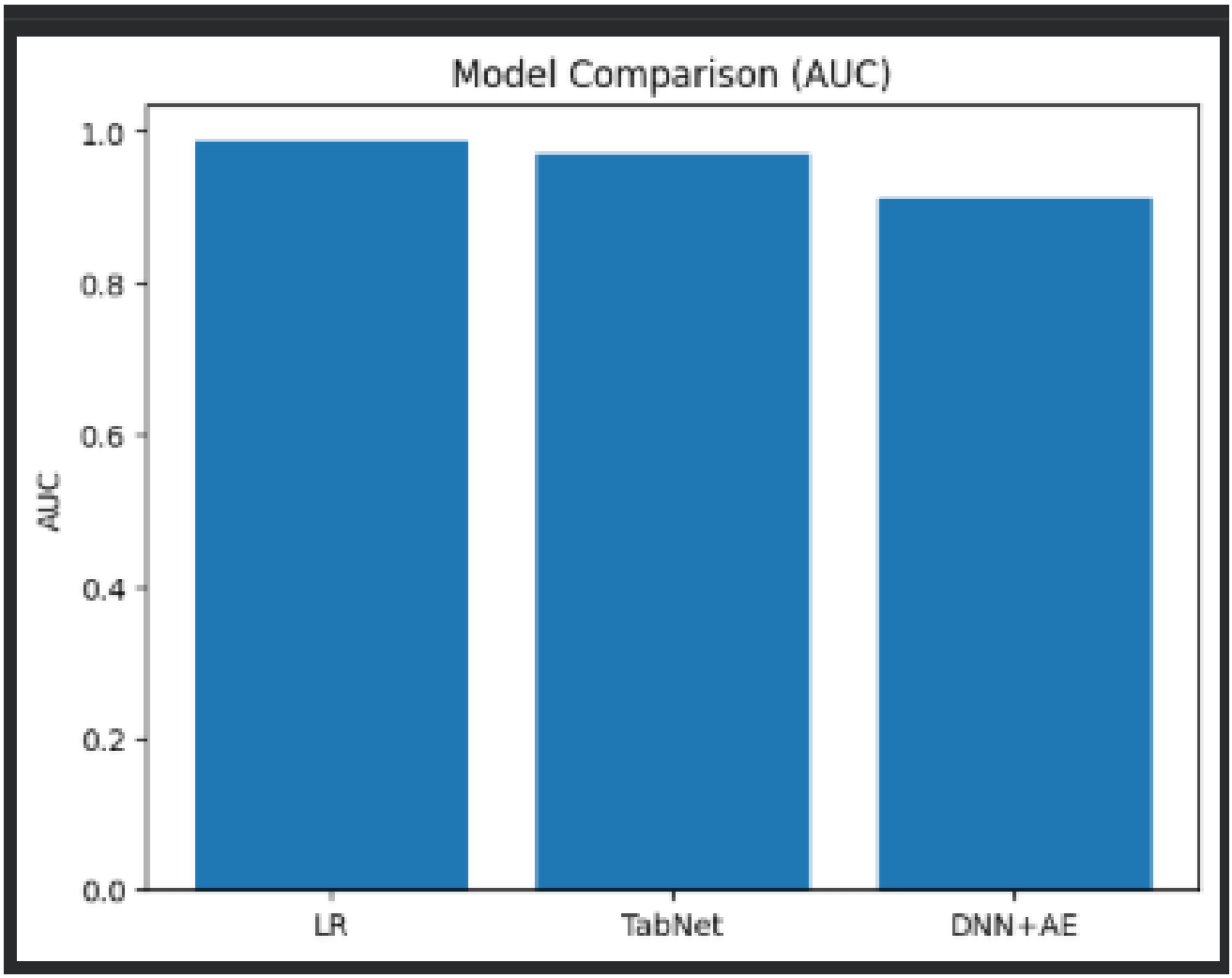
Your insight: The decline in AUC from early to later data confirms temporal drift, indicating that fraud patterns evolve over time and require periodic model retraining.



5] Comparison Graph / Table

What it shows: Performance of different models

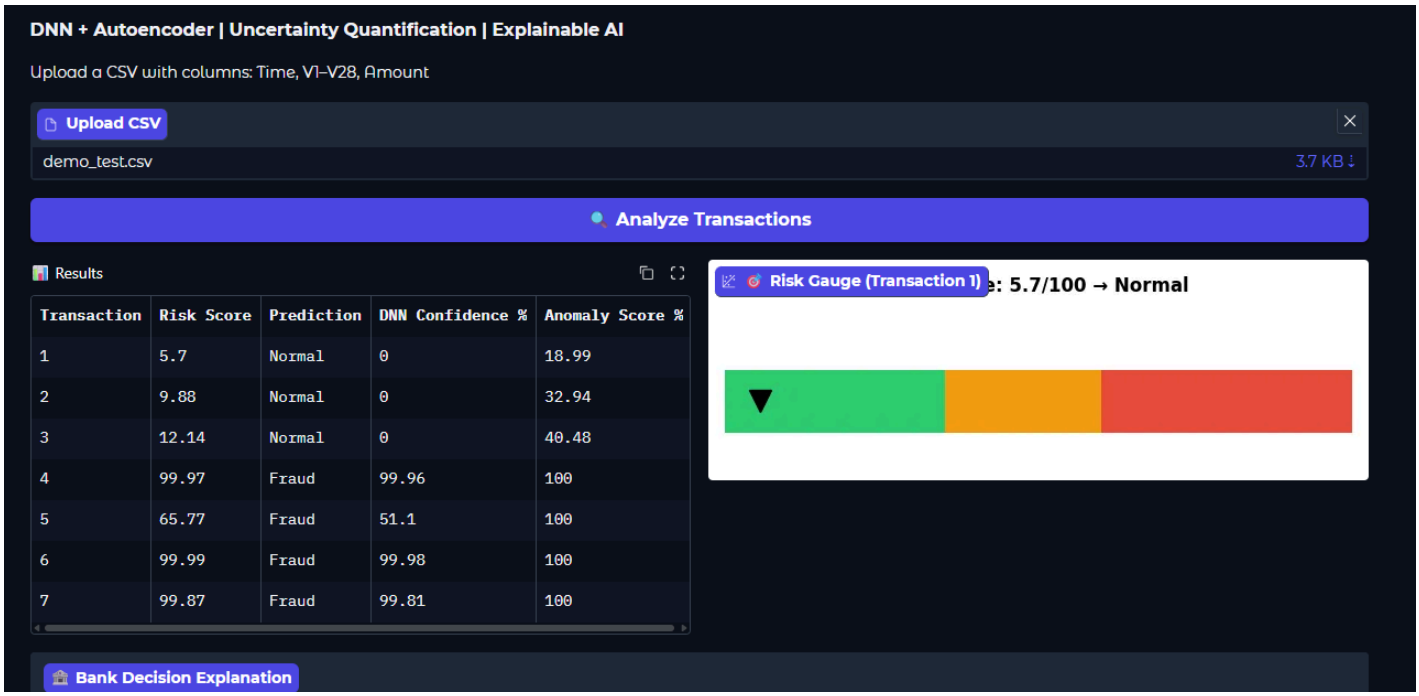
Your insight: While Logistic Regression achieves the highest AUC, the proposed system provides additional advantages such as anomaly detection, uncertainty handling, and explainability, making it more suitable for real-world deployment.



6] DNN + Autoencoder Output (Main System)

What it shows: Risk score + classification

Your insight: The dual-intelligence system combines prediction and anomaly detection to produce a more robust fraud detection mechanism, capable of identifying both known and unknown fraud patterns



Transaction	Risk Score	Prediction	DNN Confidence %	Anomaly Score %
1	5.7	Normal	0	18.99
2	9.88	Normal	0	32.94
3	12.14	Normal	0	40.48
4	99.97	Fraud	99.96	100
5	65.77	Fraud	51.1	100
6	99.99	Fraud	99.98	100
7	99.87	Fraud	99.81	100

7. Adversarial / Robustness

What it shows: Performance under manipulated inpu , Your insight: The system maintains stable performance under noisy or adversarial conditions, demonstrating improved robustness compared to single-model approaches.

```
Testing robustness...

Noise: 0.1
DNN: 81.33333333333333
System: 84.0

Noise: 0.3
DNN: 81.33333333333333
System: 84.0

Noise: 0.5
DNN: 81.33333333333333
System: 82.66666666666667

Noise: 1.0
DNN: 82.66666666666667
System: 86.66666666666667
```